

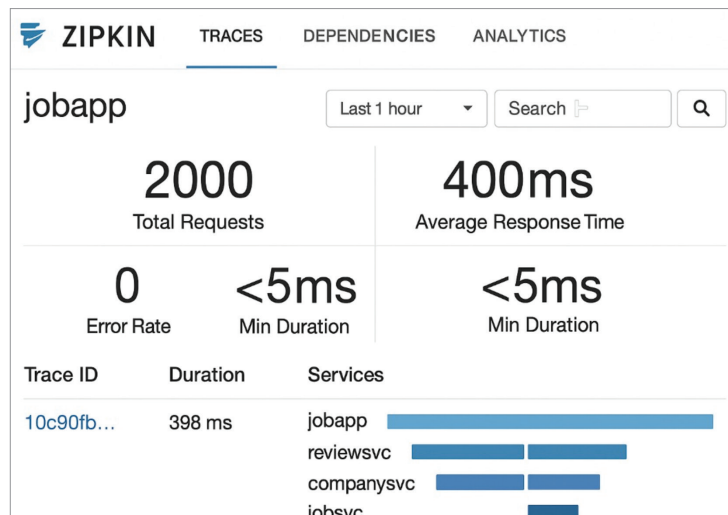


Figure 10: Metrics using Zipkin

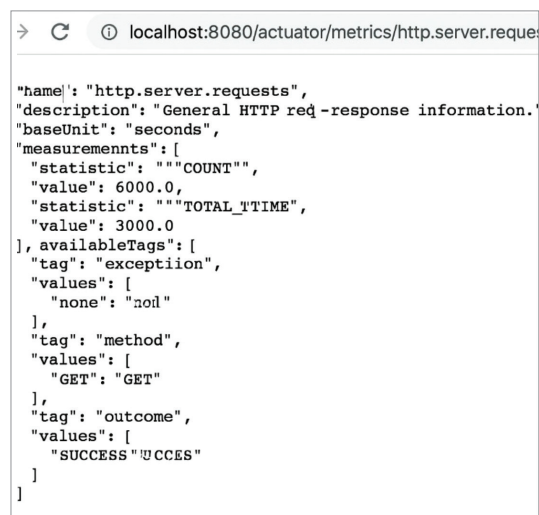


Figure 11: Metrics using Spring Boot Actuator

Quantitatively, the platform achieved a 4X improvement in scalability, 83% faster deployment, and 30–40% savings in resource utilisation, validating the architectural shift and setting a robust foundation for future expansion.

Though this migration brought significant improvements, it also introduced new complexities that warranted careful management. While scalability, resilience, and development agility were greatly enhanced, challenges such as service orchestration, distributed data management, and inter-service communication overhead emerged. To address these, techniques like asynchronous messaging (RabbitMQ), service discovery (Eureka), client-side load balancing (Ribbon), and declarative REST clients (OpenFeign) were employed.

Furthermore, the implementation of circuit breakers using Resilience4j ensured that transient failures in one service did not propagate through the system, preserving overall stability. Monitoring tools such as Zipkin for distributed tracing and Spring Boot Actuator for health checks provided critical visibility into system behaviour, enabling proactive maintenance. Overall, the microservices approach introduced manageable trade-offs, and

the net gain in system flexibility, fault isolation, and scalability validated the

architectural transition as a sustainable foundation for future expansion. **END** 🐼

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- The complete source code for this work is available at: <https://github.com/siddeshwar-kagatkar/Job-Portal>

By: Bhavya Kapadia, Siddeshwar Kagatkar and B. Thangaraju

The authors are associated with the Open Source Technology Lab at the International Institute of Technology, Bengaluru.